PHISHING DETECTION & AWARENESS REPORT

Cyber Security Task 2 By Future Interns

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About the Task

Phishing is one of the most common and costly cyber-attacks faced by businesses today. Attackers trick users into:

- clicking malicious links
- downloading infected files
- sharing passwords or OTPs

Most successful phishing attacks happen not because of weak systems, but because users don't know how to identify fake emails.

This report analyses phishing email samples, identifies red flags, and provides clear awareness guidelines for employees.

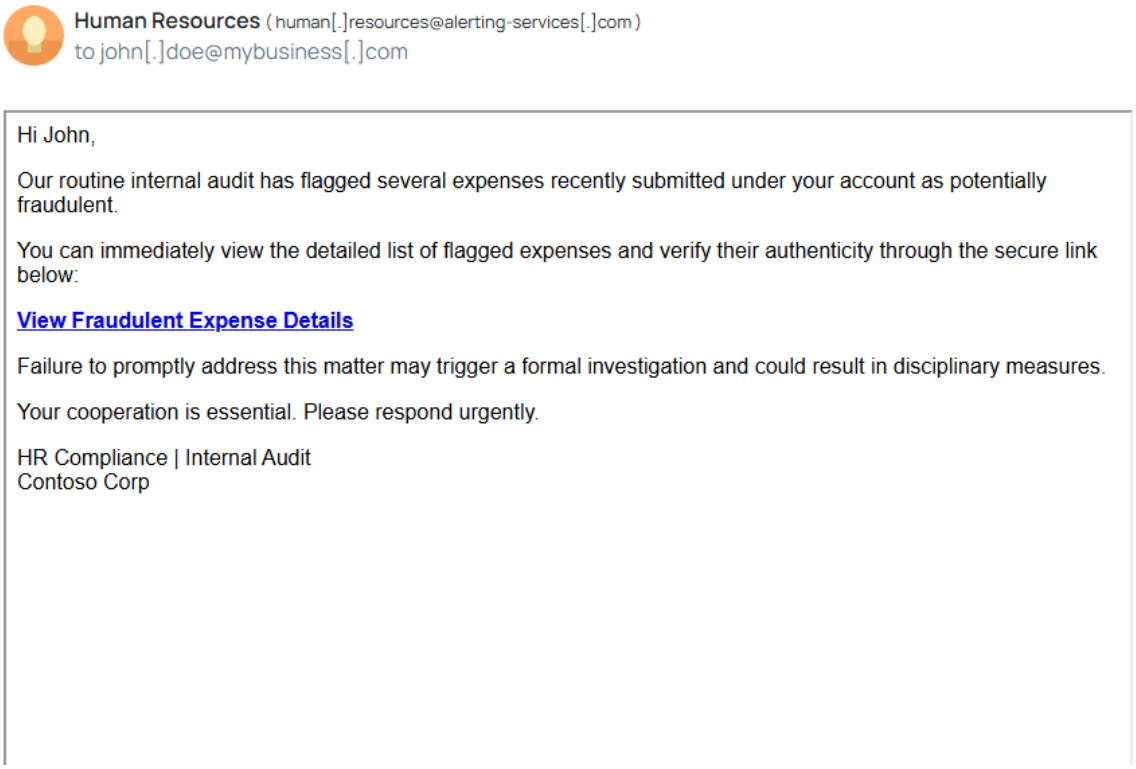
Objective

- Analyse real phishing email samples
- Identify common phishing indicators
- Classify email risk clearly
- Explain attacks in simple, non-technical language
- Create awareness guidelines to help users avoid phishing attacks

Tools Used

- **Email Header Analyser**:
    - Google Toolbox Header Analyzer
    - MXToolbox Email Header Analyzer
- **Browser Tools**: For checking links/domains safely
- **Documentation Tools**: MS Word / PDF

Sample Phishing Email (Analysed Example)



Human Resources (human[.]resources@alerting-services[.]com)
to john[.]doe@mybusiness[.]com

Hi John,

Our routine internal audit has flagged several expenses recently submitted under your account as potentially fraudulent.

You can immediately view the detailed list of flagged expenses and verify their authenticity through the secure link below:

View Fraudulent Expense Details

Failure to promptly address this matter may trigger a formal investigation and could result in disciplinary measures.

Your cooperation is essential. Please respond urgently.

HR Compliance | Internal Audit
Contoso Corp

Fig 1

Identified Red Flags

- **Suspicious sender domain**: The email comes from human.resources@alerting-services[.]com, which is not a legitimate corporate HR domain. Attackers often use lookalike or unrelated domains.
- **Urgency and fear-based language**: The message warns of "fraudulent expenses" and threatens "formal investigation and disciplinary measures" if the recipient doesn't act quickly. This is designed to pressure the user into clicking without thinking.
- **Suspicious link**: The "View Fraudulent Expense Details" link is presented as secure but points to an unknown domain. This is a classic phishing tactic to lure users into entering credentials.
- **Generic formatting and impersonation**: The email claims to be from "HR Compliance Internal Audit, Contoso Corp" but uses a generic template and mismatched sender domain, which is inconsistent with real corporate communications.
- **Targeted personalization**: The email addresses "John" directly, making it appear more convincing, but still lacks official formatting or internal references that a real HR audit would include.

Risk Classification

- **Email Type:** Phishing
- **Risk Level:** High
- **Reason:** Contains malicious link, urgency, and fake sender domain

How the Attack Works (Simple Explanation)

Attackers craft fake emails that look like they come from trusted internal departments (like HR or Finance).

1. **Deception**: The email pretends to be from HR, warning about fraudulent expenses.
2. **Urgency**: It pressures the recipient to act immediately to avoid investigation or punishment.
3. **Malicious link**: The "View Fraudulent Expense Details" button leads to a fake login page.
4. **Credential theft**: If the recipient enters their username and password, attackers capture the credentials.
5. **Exploitation**: Stolen credentials can then be used to access sensitive company systems, commit fraud, or launch further attacks.

Awareness Guidelines

Do's

- Check sender domain carefully
- Hover over links before clicking
- Report suspicious emails to IT/security team
- Use multi-factor authentication (MFA)

Don'ts

- Don't click unknown links
- Don't download unexpected attachments
- Don't share passwords or OTPs via email
- Don't trust emails with urgency or threats

References (Study Only)

- Phishing Email Examples Repository
- Phishing Mail Examples for Education
- Phishing & Non-Phishing Email Dataset
- Phishing Database (URL threats)
- Phishing Classification (ML Example)
- https://caniphish.com/free-phishing-test/phishing-email-templates (where email template on *fig 1* was soured)